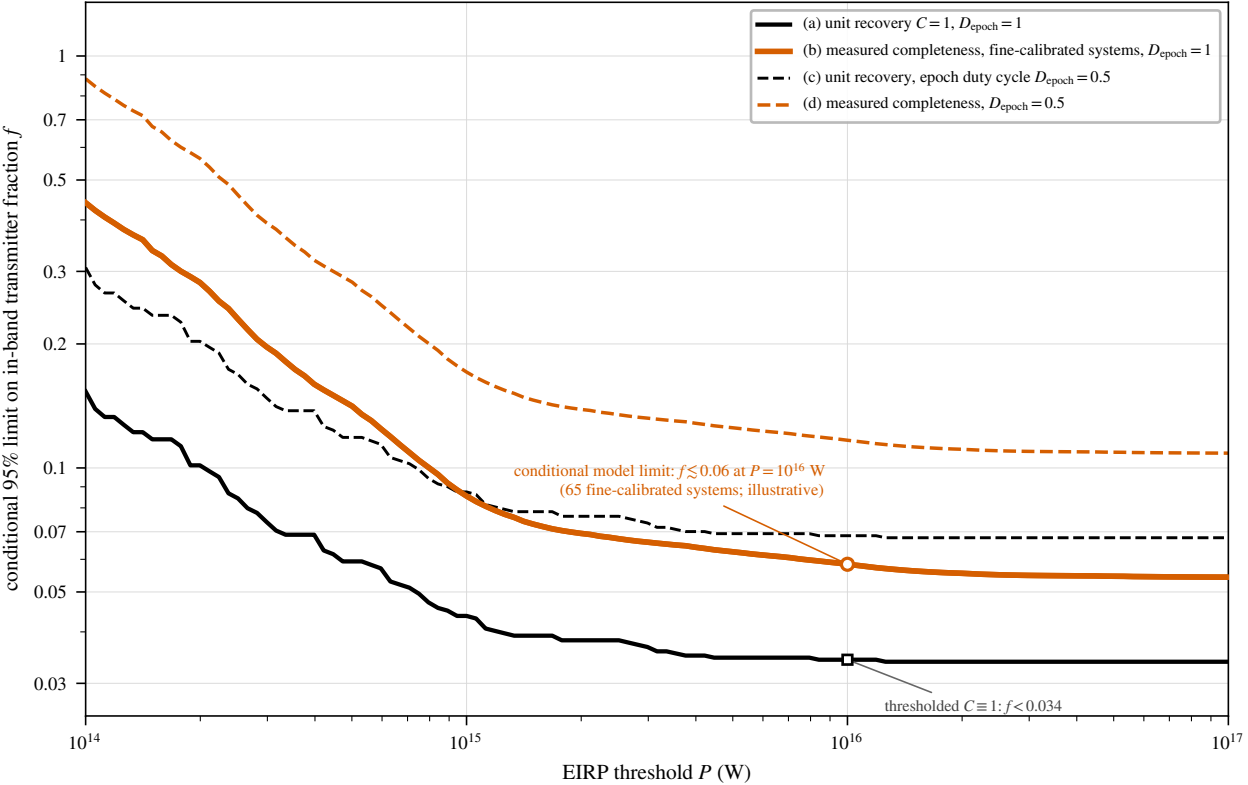




$f$  solves  $\prod_i (1 - f D_{\text{epoch}} C_i) = 0.05$ , with  $C_i$  the per-system recovery at threshold  $P$  and  $D_{\text{epoch}}$  the assumed epoch duty cycle (independent Bernoulli activity across a system's execution blocks; not the in-band transmitter fraction). A low-duty-cycle population is bounded only where  $D_{\text{epoch}} C_i$  stays large enough to exclude  $f = 1$ ; as  $D_{\text{epoch}} \rightarrow 0$  no bound exists at any  $P$ . Curve (b) is the conditional model limit quoted in the text, illustrating the model's behavior.